KFParticle Λ–^{4}He
Low / High Invariant-Mass CF Comparison

Post-processing QA report

2026-10-03 20:19 JST

1 Sample, scope and common choices

Au+Au fixed-target sample, beam-energy label 3.85 GeV; 2021 P24iy only. The beam label is not $\sqrt{s_{NN}}$. Partial merge: 26,956/26,994 subjobs; 38 excluded or missing. Input: `auau3p85_anaFemtoLambda_4He_KFParticle_2021P24iy_partial_merge.root`. No event reanalysis, selection-cut change or original ROOT modification. k^* bin width: 20 MeV/c; display: 0–0.3 GeV/c; normalization: 0.4–0.6 GeV/c. The normalization interval remains in use even outside the displayed range. The integrated comparison uses $0 \leq k^* < 0.2$ GeV/c. It is the normalized ratio of summed counts, not an arithmetic average of CF points. Each mass window is independently normalized.

2 Inclusive 0–80% comparison

Ratios compare each half to full within the same background method. NA means unavailable, not zero or agreement. Errors are approximate (see limitations below).

- Raw: low/full = 1.001 ± 0.00606; high/full = 0.9992 ± 0.00748.
- SBsub: low/full = 1.001 ± 0.00644; high/full = 0.9984 ± 0.00805.
- CFsub: low/full = 1.001 ± 0.00625; high/full = 0.9986 ± 0.0078.
- Method 3: low/full = NA; high/full = NA.
- Method 5: low/full = 1 ± 0.00607; high/full = 0.9998 ± 0.00803.

3 Mass windows and methods

Exact gates use candidate masses. Mass-fit projections and inclusive purity use whole stored mass bins, without split-bin interpolation:

Window	Exact nominal edges	Whole-bin edges
full	1.10708–1.12484	1.107–1.125
low	1.10708–1.11596	1.107–1.116
high	1.11596–1.12484	1.116–1.125
leftSB	1.08044–1.10708	1.08–1.107
rightSB	1.12484–1.15148	1.125–1.151

Edges are in GeV/c^2. Exact low excludes the mean, exact high includes it; the full outer edges are included. Whole-bin intervals are disjoint and half-open. Raw/SBsub/CFsub signal histograms use exact gates, while Methods 3/5 include this documented mass-binning difference.

Raw Normalized signal SE/ME, without background correction.

SBsub Subtract summed sideband counts from SE and ME before normalization, using 1/3 for full and 1/6 for each half. A locally flat background is assumed.

CFsub $C_r^{\mathrm{corr}} = [C_r - (1 - P_r)C_{\mathrm{SB}}]/P_r$. Each window uses its own current inclusive purity proxy.

Method 3 Fit each SE/ME mass projection with Gaussian plus quadratic, then form the normalized ratio of $N - B$. Low/high background integrals share one fit; full uses their sum.

Method 5 Purity fits use 50 MeV/c coarse bins and $W = 2$ neighbors per side: interior window 250 MeV/c, truncated at edges. Each original 5 MeV/c count receives the purity of its containing coarse bin, then contributes to the 20 MeV/c CF bin. No fine-bin refit is performed. Full has its own window purity; its corrected yield need not equal the sum of corrected half yields.

Window	Inclusive purity proxy	Actual width factor
full	$0.9672 \pm 2.14e-05$	0.3396226
low	$0.9716 \pm 1.91e-05$	0.1698113
high	$0.9612 \pm 2.68e-05$	0.1698113

These purity errors are not propagated into CFsub. These are inclusive proxies, not measured pair-, centrality- or momentum-dependent purities; sideband asymmetry and background-model systematics remain.

4 Integrated CF values and half/full ratios

Valid bins lists full/low/high counts in the displayed interval. A partial curve does not guarantee that the complete integration interval is valid. Invalid bins are not silently removed from the requested integration.

Method	C_{full}	C_{low}	C_{high}	Low/Full	High/Full	Valid bins
Centrality 0-80%						
Raw	1.056 ± 0.0071	1.057 ± 0.00956	1.055 ± 0.0106	1.001 ± 0.00606	0.9992 ± 0.00748	15/15/15 of 15
SBsub	1.053 ± 0.00765	1.054 ± 0.0102	1.051 ± 0.0115	1.001 ± 0.00644	0.9984 ± 0.00805	15/15/15 of 15
CFsub	1.053 ± 0.00737	1.054 ± 0.00985	1.051 ± 0.0111	1.001 ± 0.00625	0.9986 ± 0.0078	15/15/15 of 15
Method 3	NA	1.065 ± 0.0101	NA	NA	NA	14/15/14 of 15
Method 5	1.054 ± 0.00736	1.055 ± 0.00972	1.054 ± 0.0113	1 ± 0.00607	0.9998 ± 0.00803	15/15/15 of 15
Centrality 20-80%						
Raw	1.126 ± 0.0231	1.13 ± 0.0315	1.121 ± 0.0341	1.004 ± 0.0189	0.9958 ± 0.0223	15/15/15 of 15
SBsub	1.144 ± 0.0269	1.148 ± 0.0359	1.139 ± 0.0398	1.004 ± 0.0211	0.9956 ± 0.0254	15/15/15 of 15
CFsub	1.125 ± 0.024	1.13 ± 0.0324	1.12 ± 0.0355	1.004 ± 0.0195	0.9955 ± 0.0232	15/15/15 of 15
Method 3	NA	NA	NA	NA	NA	13/13/13 of 15
Method 5	1.162 ± 0.0251	1.169 ± 0.0335	1.154 ± 0.0379	1.005 ± 0.0192	0.9925 ± 0.0243	15/15/15 of 15
Centrality 0-20%						
Raw	1.049 ± 0.00747	1.05 ± 0.01	1.048 ± 0.0112	1.001 ± 0.0064	0.9993 ± 0.00794	15/15/15 of 15
SBsub	1.045 ± 0.00799	1.046 ± 0.0106	1.043 ± 0.012	1.001 ± 0.00676	0.9986 ± 0.0085	15/15/15 of 15
CFsub	1.046 ± 0.00775	1.047 ± 0.0104	1.044 ± 0.0117	1.001 ± 0.0066	0.9988 ± 0.00828	15/15/15 of 15
Method 3	NA	1.057 ± 0.0106	NA	NA	NA	14/15/14 of 15
Method 5	1.046 ± 0.00771	1.046 ± 0.0102	1.047 ± 0.0118	0.9997 ± 0.00639	1.001 ± 0.0085	15/15/15 of 15
Centrality 0-10%						
Raw	1.046 ± 0.00827	1.047 ± 0.0111	1.045 ± 0.0124	1.001 ± 0.00709	0.999 ± 0.00882	15/15/15 of 15
SBsub	1.042 ± 0.0088	1.043 ± 0.0117	1.04 ± 0.0132	1.002 ± 0.00747	0.9982 ± 0.00941	15/15/15 of 15
CFsub	1.043 ± 0.00859	1.045 ± 0.0115	1.041 ± 0.0129	1.001 ± 0.00732	0.9984 ± 0.00921	15/15/15 of 15
Method 3	NA	NA	NA	NA	NA	14/14/14 of 15
Method 5	1.043 ± 0.00851	1.043 ± 0.0112	1.045 ± 0.0131	0.9993 ± 0.00707	1.001 ± 0.00943	15/15/15 of 15
Centrality 10-20%						
Raw	1.076 ± 0.0177	1.079 ± 0.0239	1.074 ± 0.0264	1.002 ± 0.0149	0.9982 ± 0.0182	15/15/15 of 15
SBsub	1.08 ± 0.0194	1.083 ± 0.0259	1.078 ± 0.029	1.002 ± 0.016	0.9977 ± 0.0198	15/15/15 of 15
CFsub	1.075 ± 0.0184	1.078 ± 0.0247	1.073 ± 0.0275	1.002 ± 0.0154	0.9979 ± 0.0189	15/15/15 of 15
Method 3	NA	NA	NA	NA	NA	13/13/13 of 15
Method 5	1.087 ± 0.0186	1.091 ± 0.0246	1.084 ± 0.0283	1.003 ± 0.0149	0.997 ± 0.0195	15/15/15 of 15
Centrality 20-60%						
Raw	1.126 ± 0.0232	1.132 ± 0.0315	1.121 ± 0.0341	1.005 ± 0.0189	0.9949 ± 0.0223	15/15/15 of 15
SBsub	1.144 ± 0.0269	1.15 ± 0.036	1.138 ± 0.0398	1.005 ± 0.0211	0.9946 ± 0.0254	15/15/15 of 15
CFsub	1.126 ± 0.024	1.131 ± 0.0325	1.12 ± 0.0355	1.005 ± 0.0195	0.9946 ± 0.0232	15/15/15 of 15
Method 3	NA	NA	NA	NA	NA	13/13/13 of 15
Method 5	1.155 ± 0.025	1.161 ± 0.0334	1.146 ± 0.0378	1.006 ± 0.0192	0.9922 ± 0.0244	15/15/15 of 15

5 Validity and uncertainty limitations

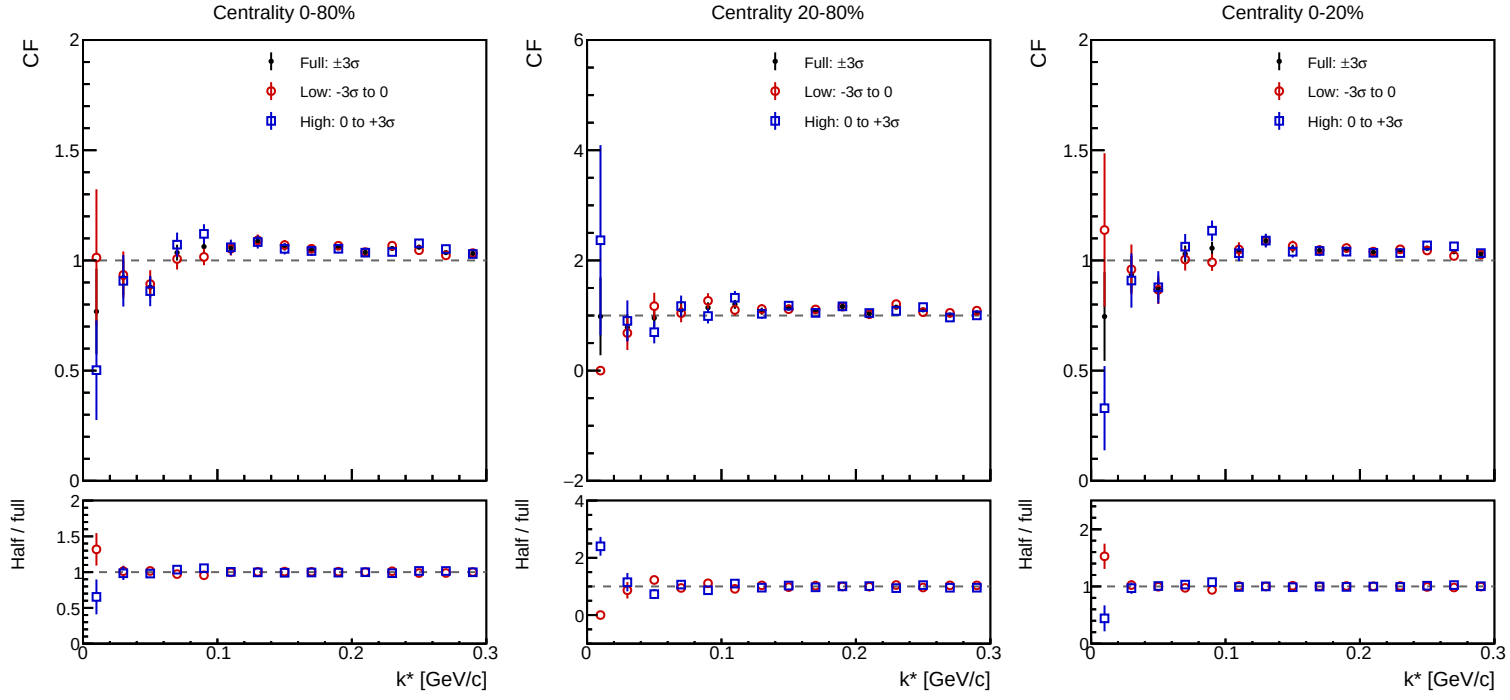
An invalid fit in a required normalization bin makes its entire CF unavailable. Invalid comparison bins make the integrated contrast NA, not zero. Numerical convergence does not prove mass-model adequacy: large chi-square/ndf or small fit probabilities prohibit precision-model claims. Detailed fit diagnostics and NA reasons are in the Japanese README and extraction JSON. First-order propagation retains shared full/half counts, sidebands, normalization integrals, the low/high background covariance from each common fit, and reuse of one coarse-bin purity by multiple CF bins. It cannot recover event / track sharing or mixed-pair reuse correlations. Covariance between observed N and fitted background B, covariance between overlapping fits in different coarse bins, CFsub purity uncertainty and model systematics are not included. These comparisons are QA contrasts, not final significance tests. Centrality groups overlap and are not independent. Low/high CFs cannot be directly added. Small TH2F rounding residuals are recorded without silently replacing the full histogram.

6 Figure guide

The next ten pages show two pages per method: Raw, SBsub, CFsub, Method 3, Method 5. First page: 0–80%, 20–80%, 0–20%; second: 0–10%, 10–20%, 20–60%. Upper panels: full (black), low (red), high (blue). Lower panels: low/full and high/full with supplied covariance errors. Failed bins are omitted, not displayed as measured zero.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Raw SE / ME

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

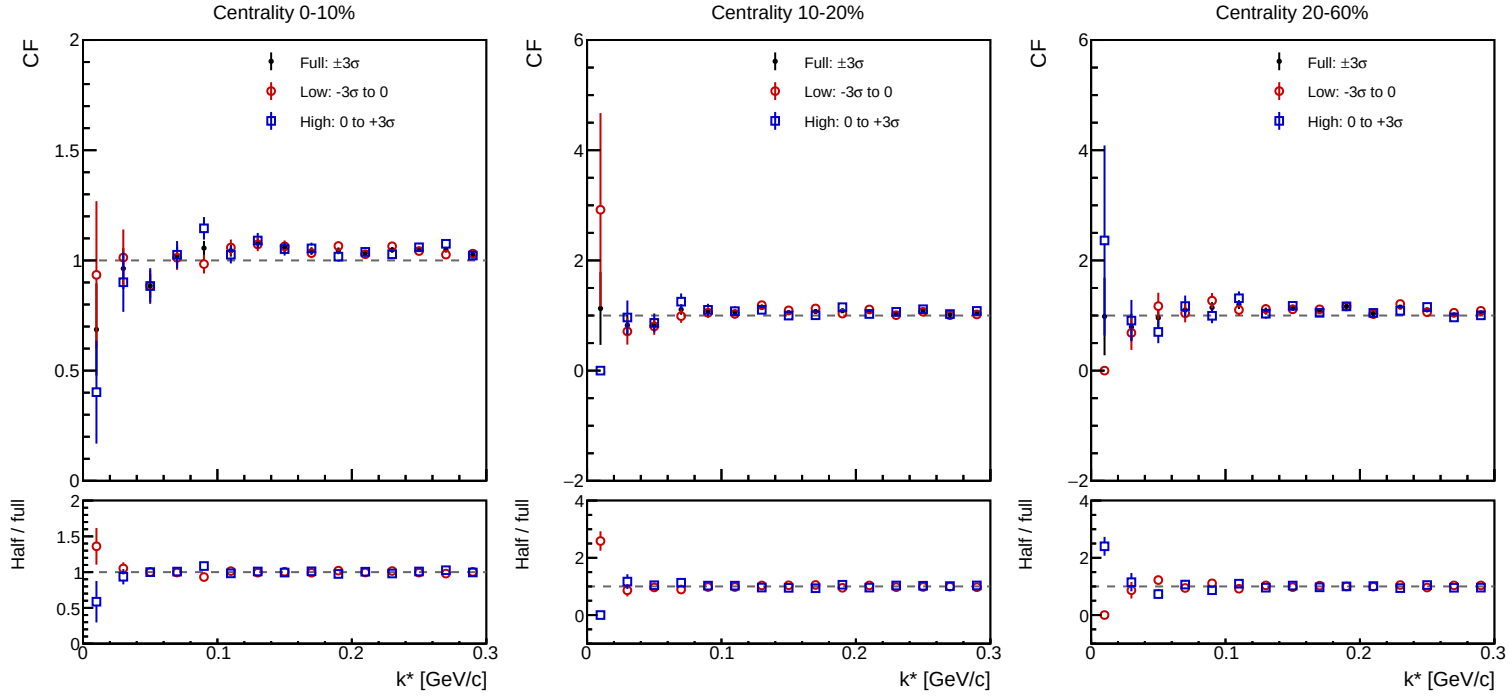


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Raw SE / ME

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

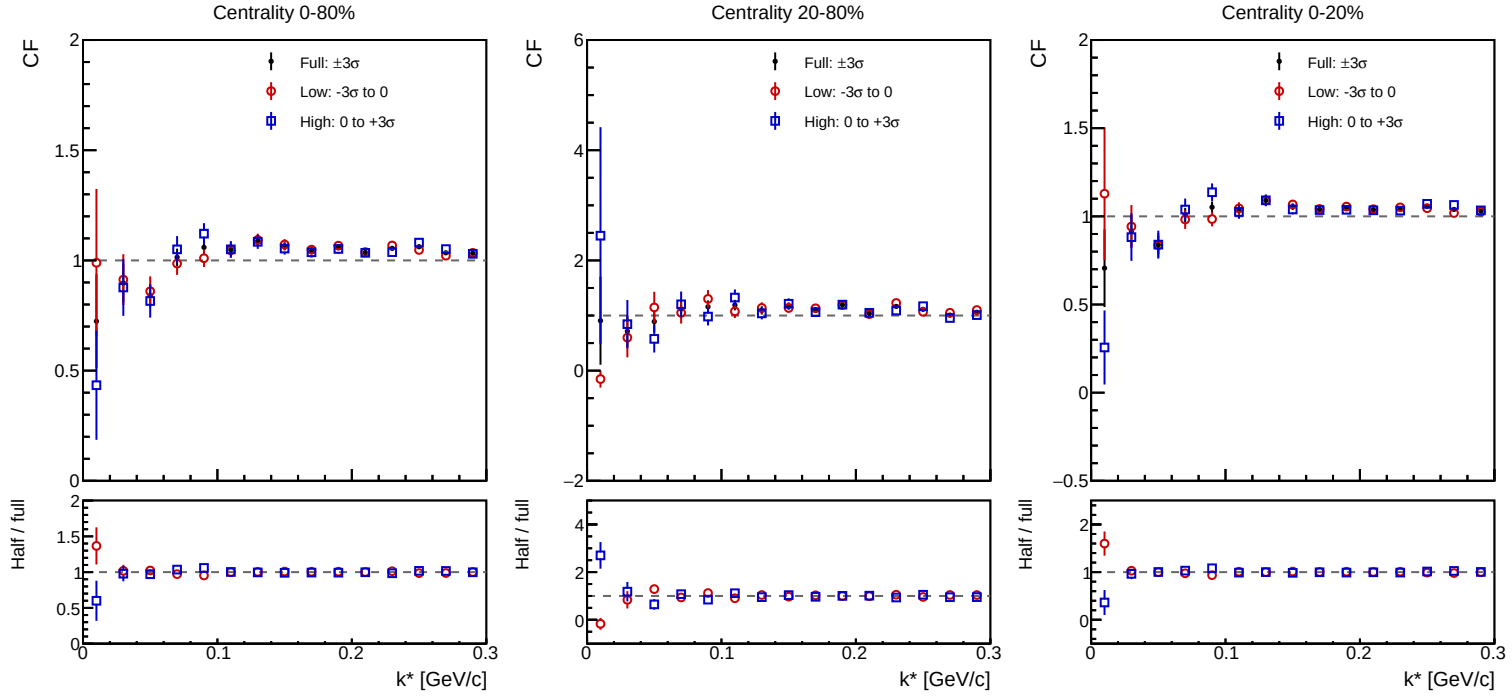


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Count-level sideband subtraction

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

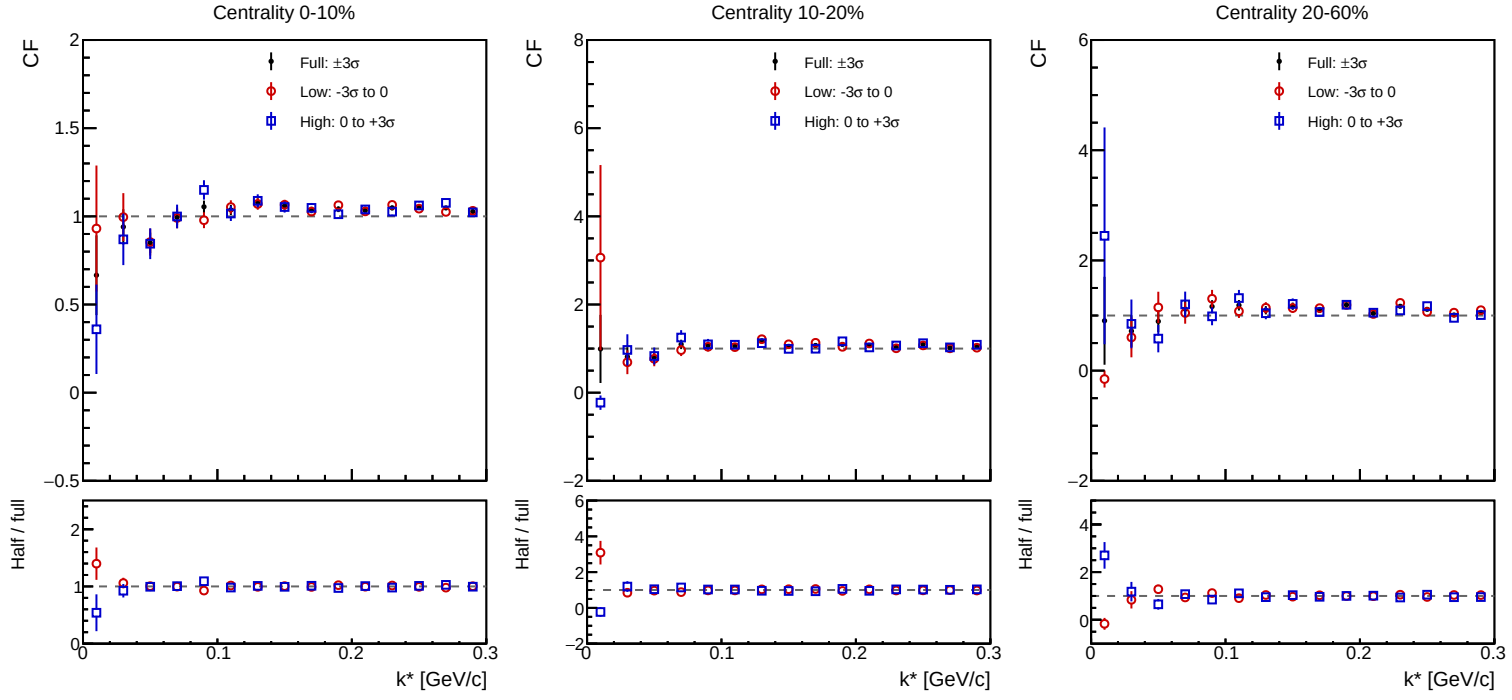


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Count-level sideband subtraction

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

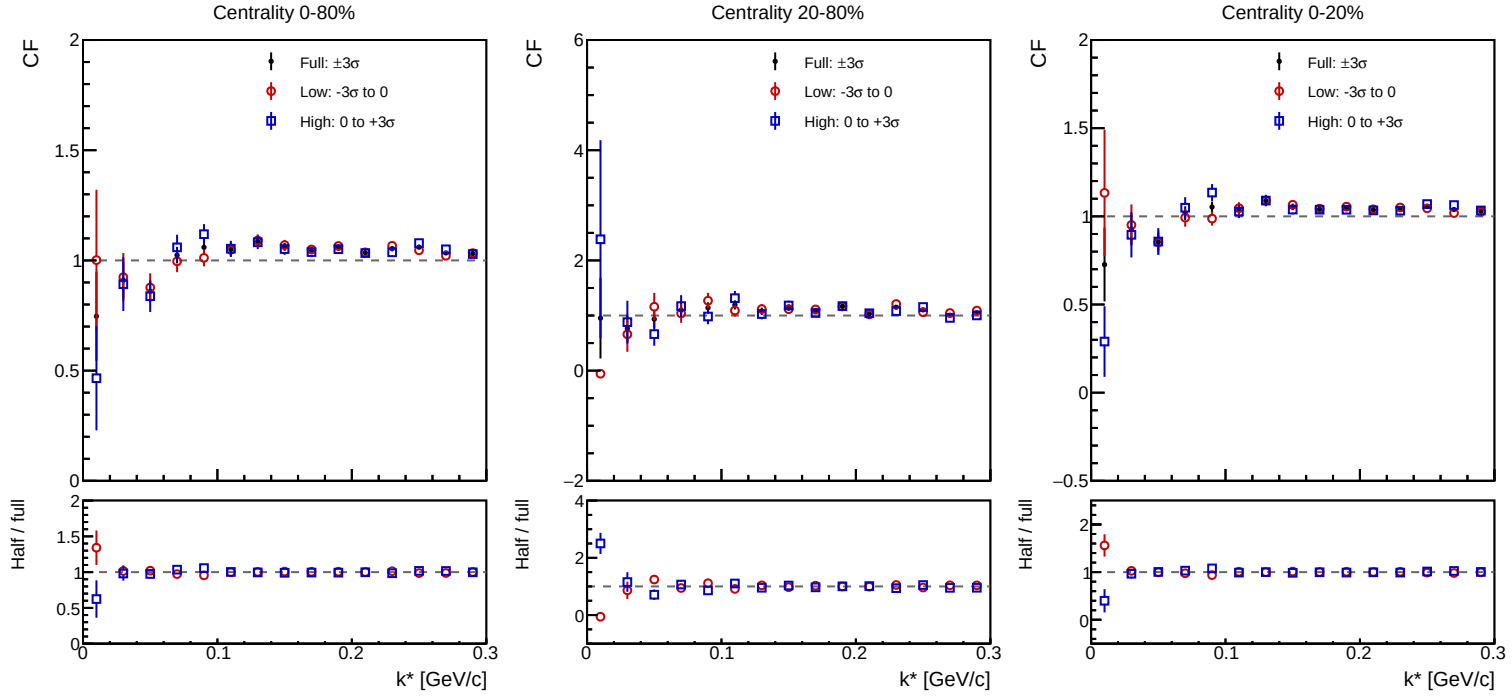


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | CF-level purity correction

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

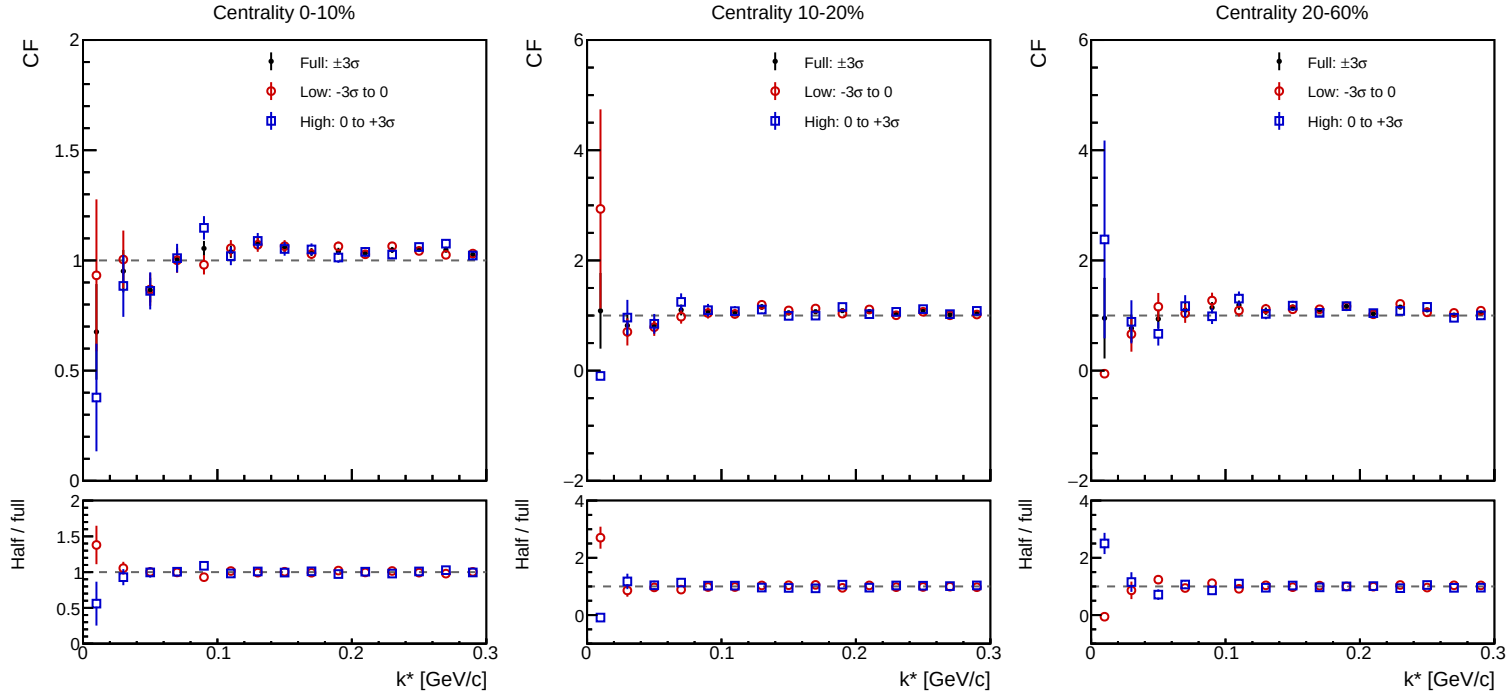


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | CF-level purity correction

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

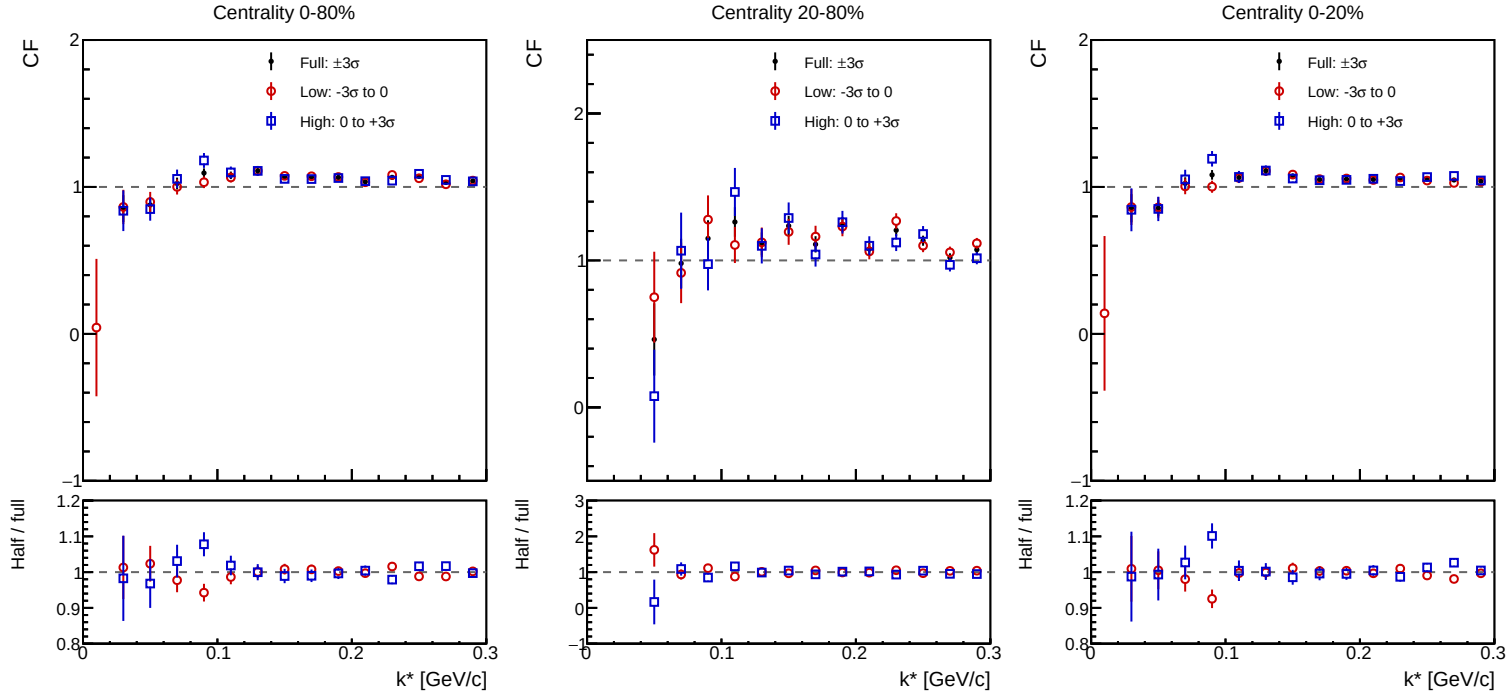


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Independent k* mass fits

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

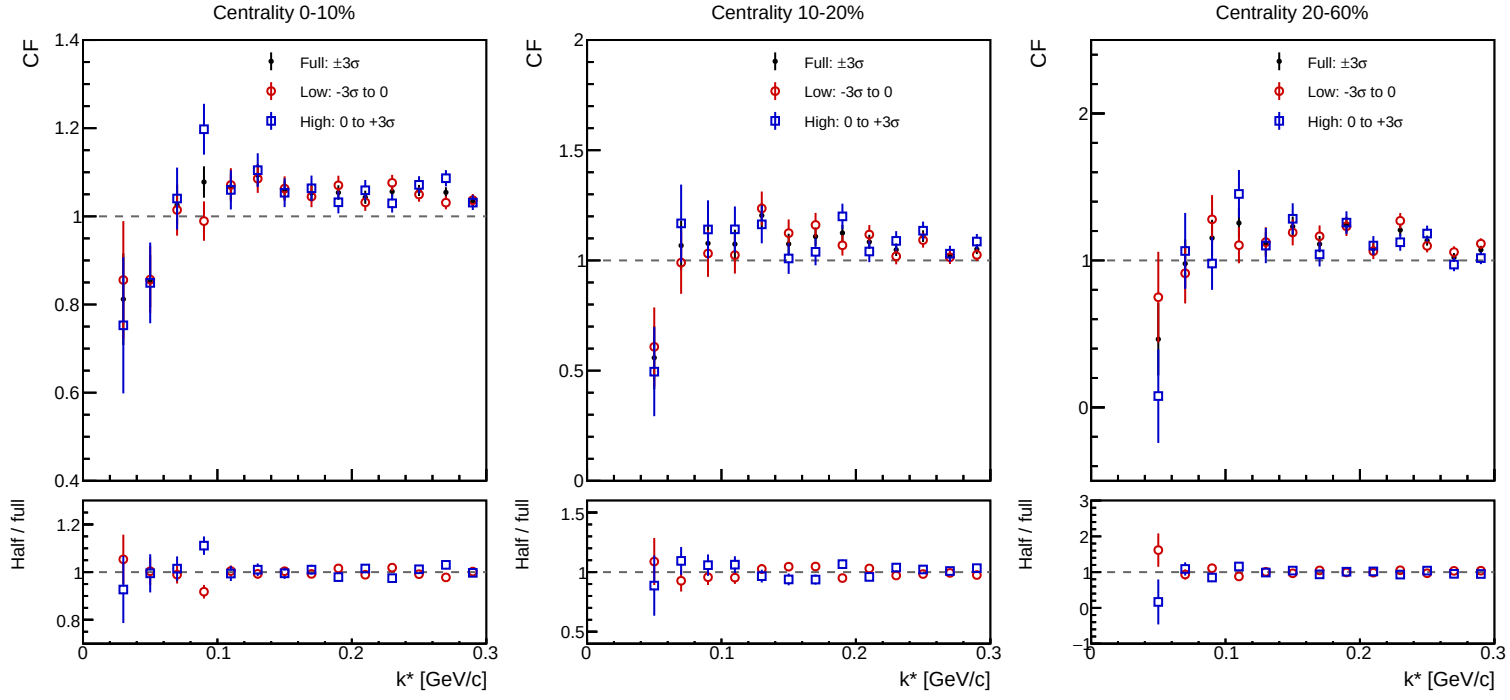


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Independent k* mass fits

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

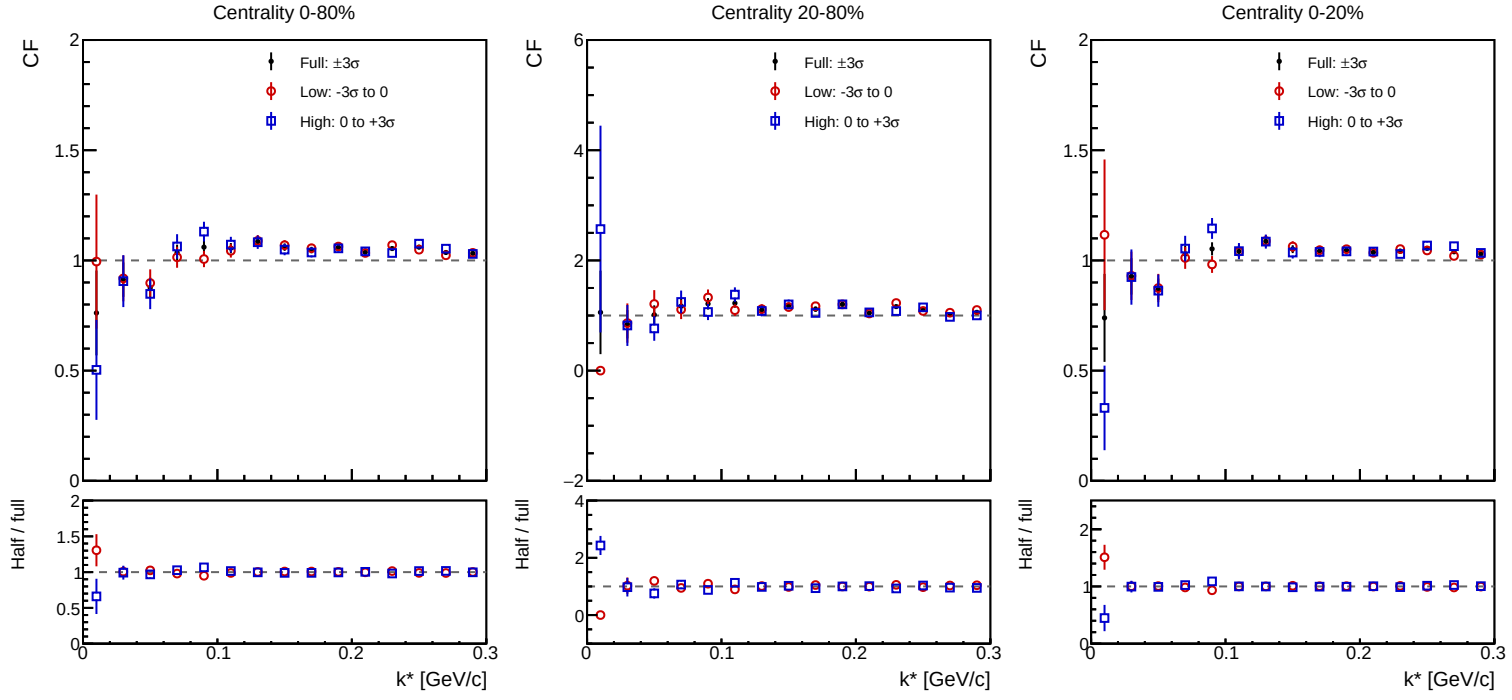


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Sliding-window mass fits

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c

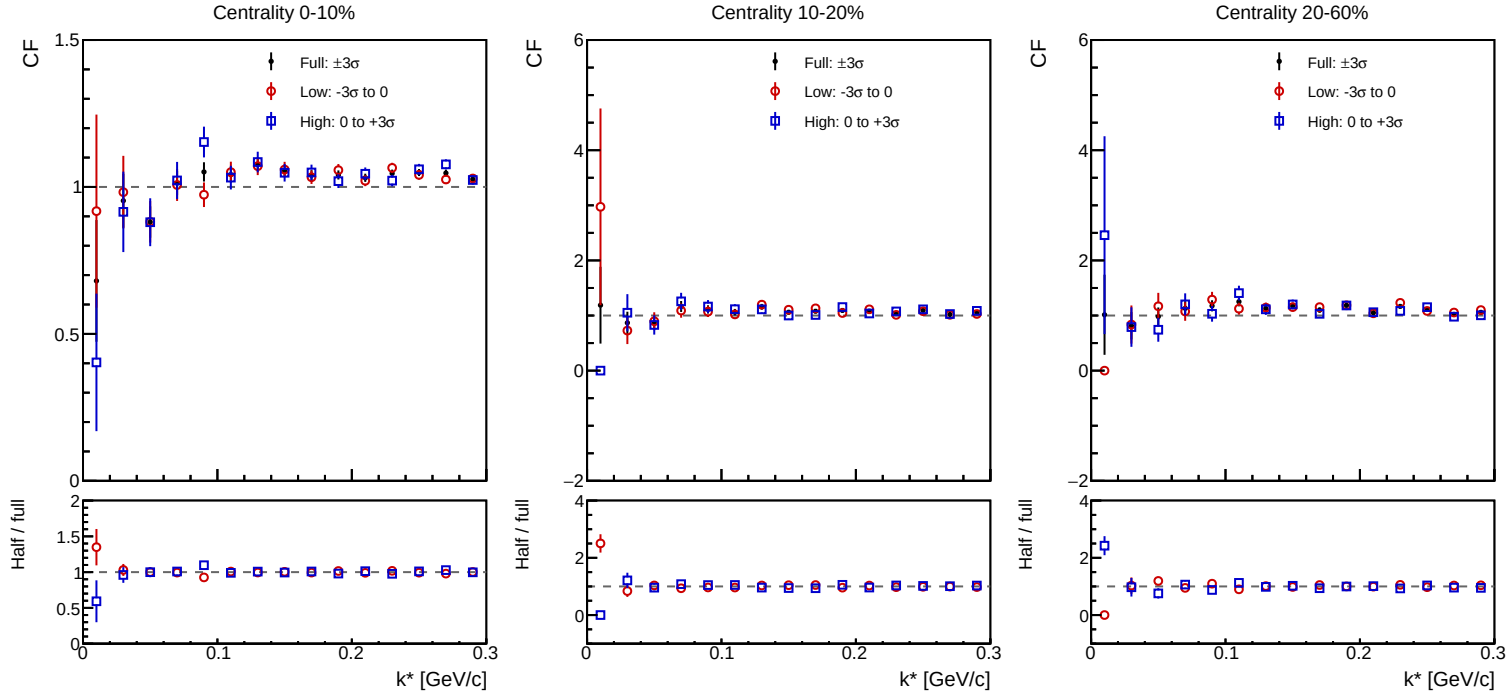


Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.

Au+Au 3.85 GeV, KFParticle | Λ-^{4}He | Sliding-window mass fits

Mass-window comparison; each CF independently normalized in 0.40-0.60 GeV/c



Normalization range shaded where visible. Lower panels: low/full (red), high/full (blue).

Invalid / failed bins omitted; ratios use supplied covariance errors.